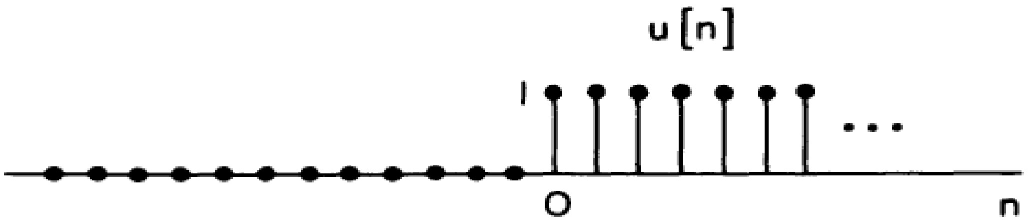
A Review on Discrete-Time Signals

Important Discrete Signals

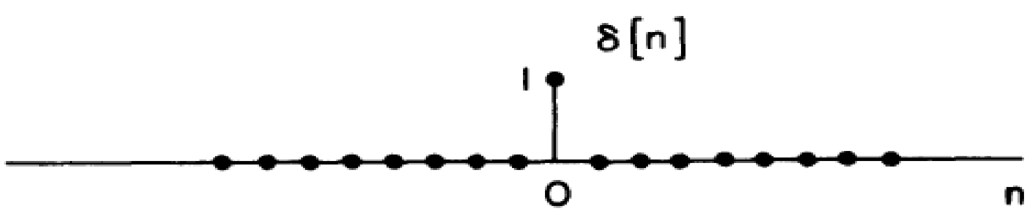
UNIT STEP FUNCTION: DISCRETE-TIME



$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

UNIT IMPULSE FUNCTION: DISCRETE-TIME

(Unit Sample)



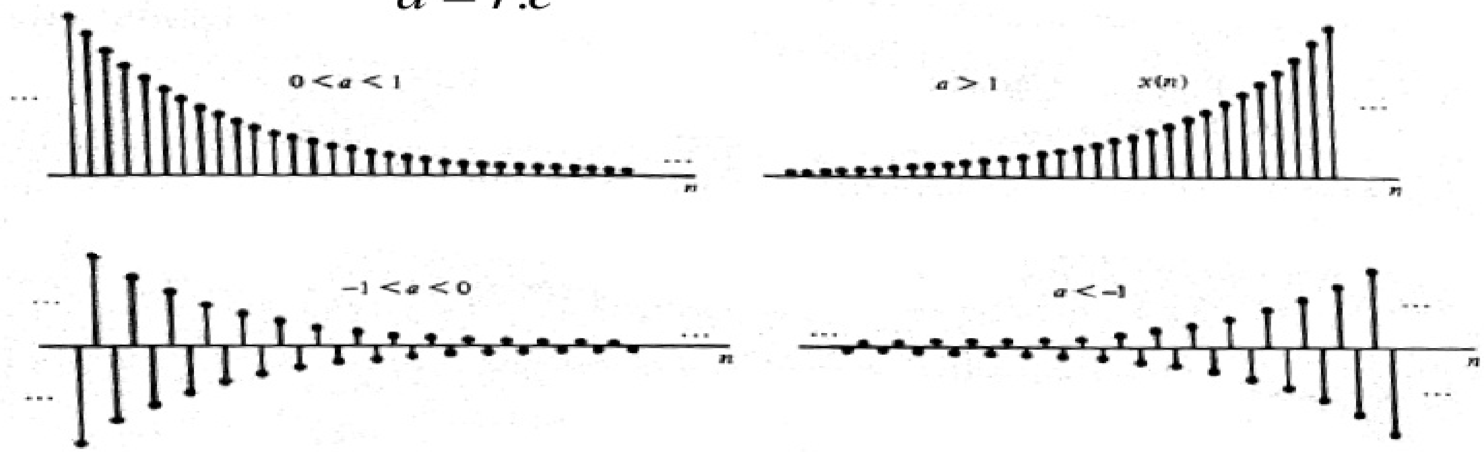
$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

Important Discrete Signals

exponential signal:

$$x(n) = a^n \qquad for \ \ \forall n$$

$$a = r.e^{j\theta}$$



When |a|>1 The signal converges to 0 at ∞

Basic Operations on Discrete Signals

Shifting This is the transformation defined by $f(n) = n - n_0$. If $y(n) = x(n - n_0)$, $x(n)$ is shifted to the right by n_0 samples if n_0 is positive (this is referred to as a delay), and it is shifted to the left by n_0 samples if n_0 is negative (referred to as an advance).

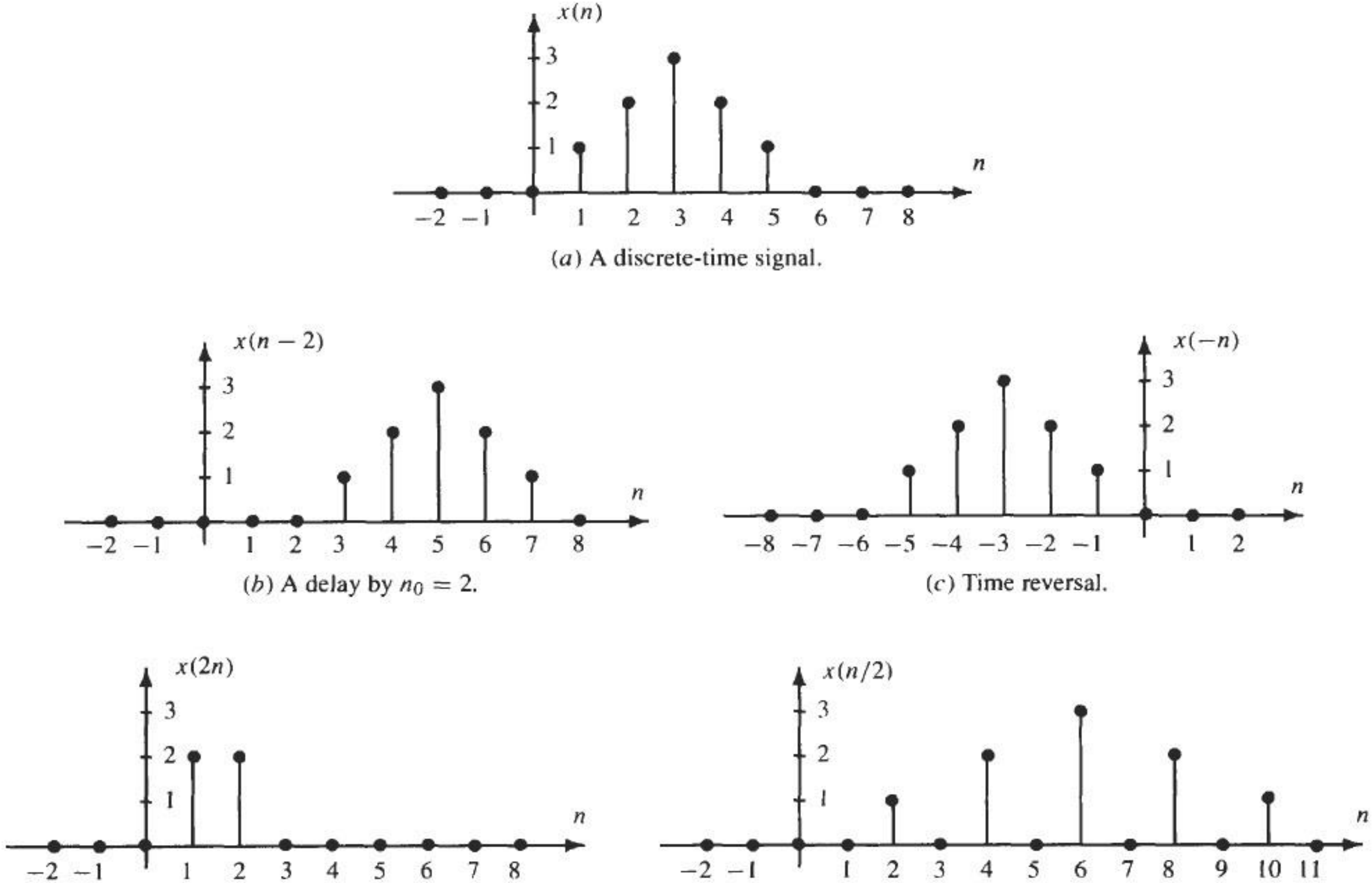
Reversal This transformation is given by $f(n) = -n$ and simply involves "flipping" the signal $x(n)$ with respect to the index n.

Time Scaling This transformation is defined by $f(n) = Mn$ or $f(n) = n/N$ where M and N are positive integers. In the case of $f(n) = Mn$, the sequence $x(Mn)$ is formed by taking every Mth sample of $x(n)$ (this operation is known as *down-sampling*). With $f(n) = n/N$ the sequence $y(n) = x(f(n))$ is defined as follows:

$$y(n) = \begin{cases} x\left(\dfrac{n}{N}\right) & n = 0, \pm N, \pm 2N, \cdots \\ 0 & \text{otherwise} \end{cases}$$

(this operation is known as *up-sampling*).

Basic Operations on Discrete Signals



(*a*) A discrete-time signal.

(*b*) A delay by $n_0 = 2$.

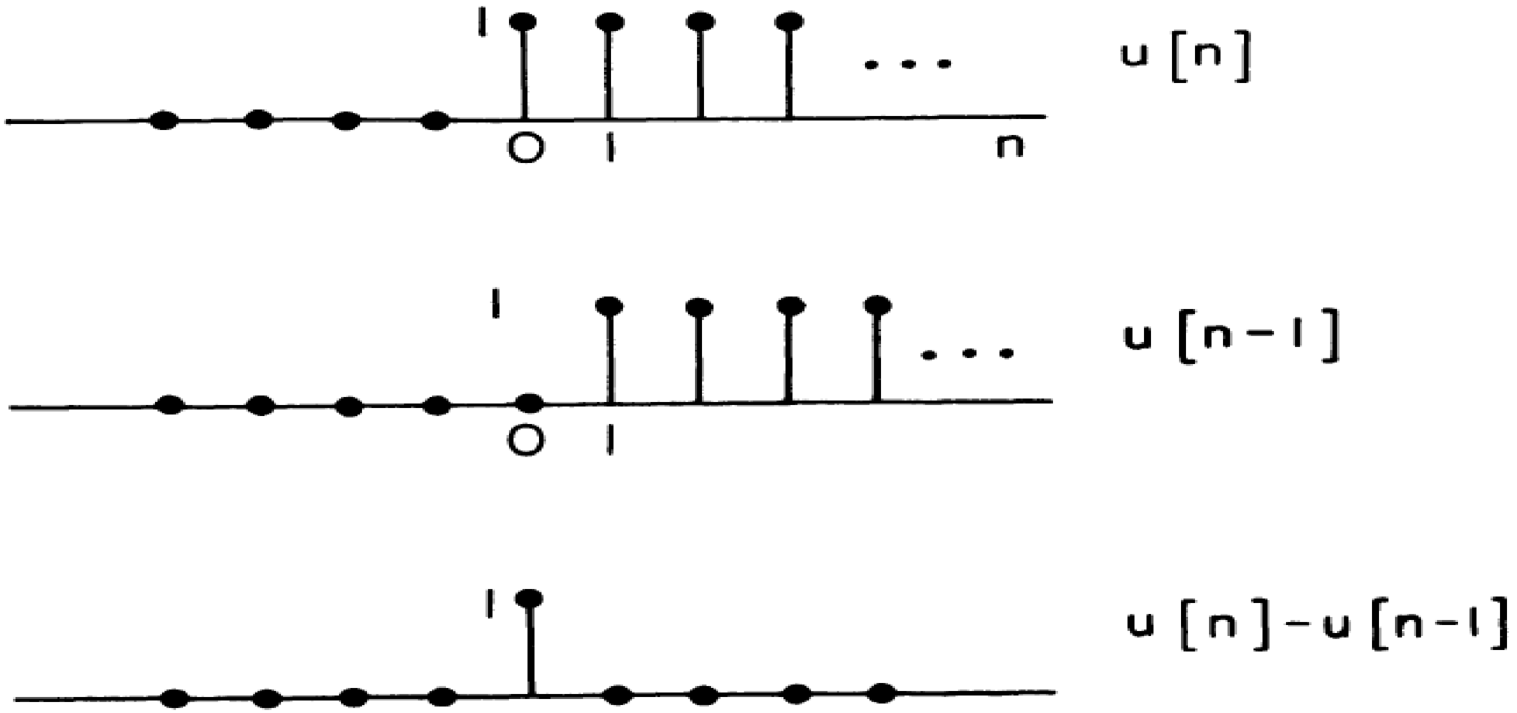
(*c*) Time reversal.

(*d*) Down-sampling by a factor of 2.

(*e*) Up-sampling by a factor of 2.

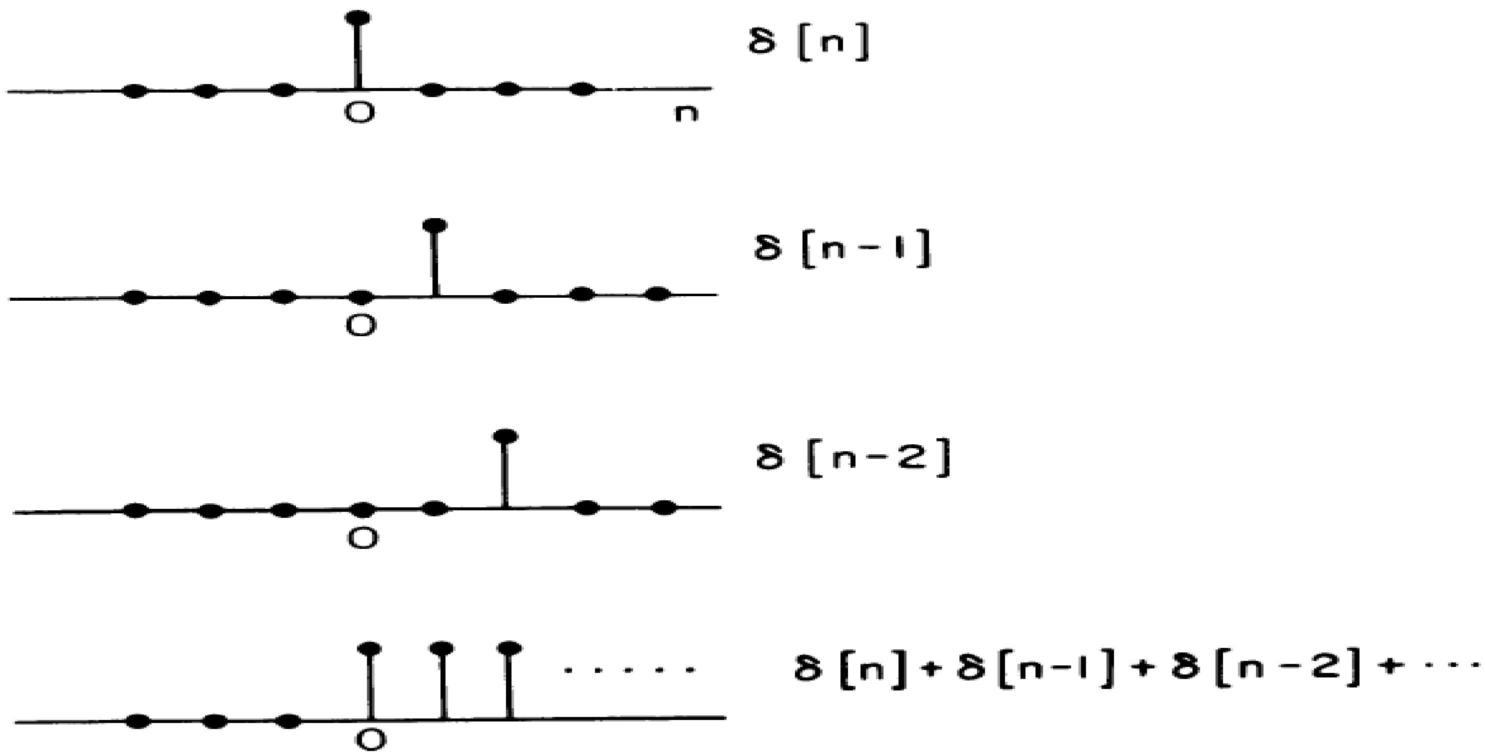
Relation Between Unit Impulse and Unit Step Functions

$$\delta[n] = u[n] - u[n-1]$$

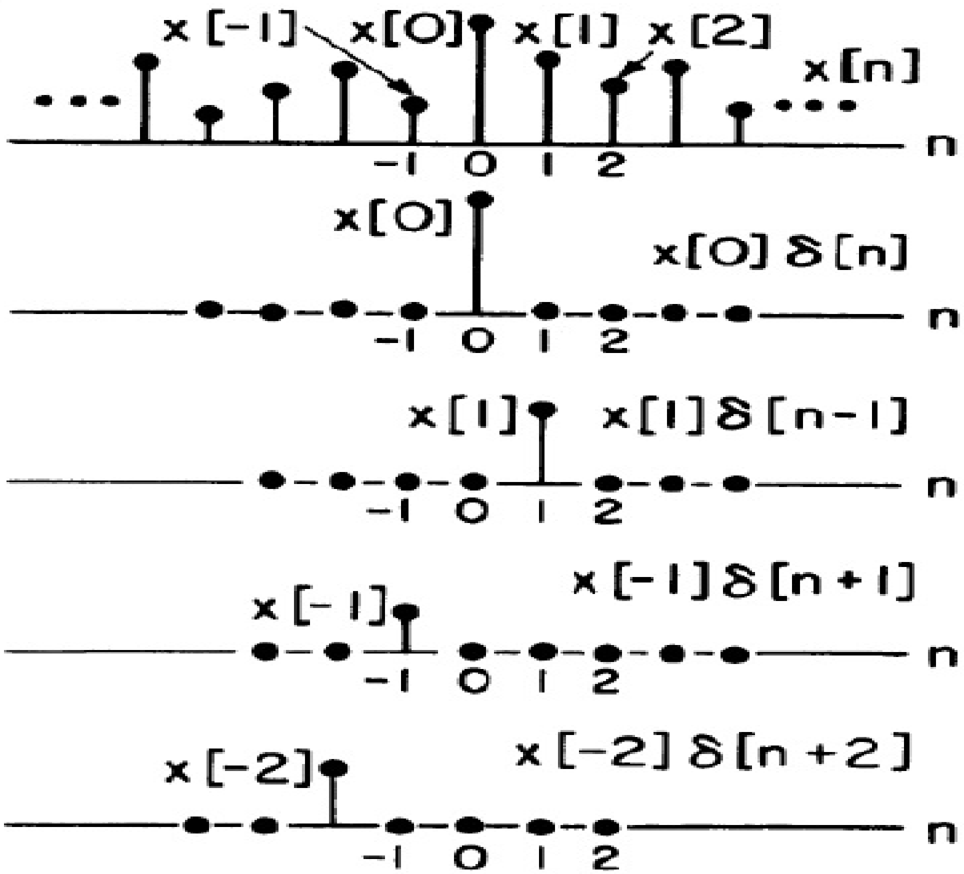


Relation Between Unit Impulse and Unit Step Functions

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k]$$



Relation Between any Signal and Unit Impulse Function



$$x[n] =$$

$$x[0]\delta[n] + x[1]\delta[n-1$$

$$+ x[-1]\delta[n+1] + \cdots$$

$$= \sum_{k=-\infty}^{+\infty} x[k]\delta[n-k]$$

A Very Important Relation

$$x(n) = \sum_{k=-\infty}^{+\infty} x(k)\delta(n-k)$$